

A PROJECT TITLE
ON
Sports & Entertainment: Match Outcome Prediction Using Logistic Regression and Random Forest

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY
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by

Jakka Bhanu
V. Charishma

Roll no - 2520030006
Roll no - 2520030140

Under the Esteemed guidance of

Shaik Asif

Assistant Professor

CSE



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in



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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: Sports & Entertainment: Match Outcome Prediction Using Logistic Regression and Random Forest

1. ABSTRCAT:

Predicting the outcome of sporting events has become an important application area within the sports and entertainment industry, driven by growing interest from fans, broadcasters, team management, and the sports betting sector. This project presents a machine learning based approach to predict match outcomes using two well established classification algorithms: Logistic Regression and Random Forest. Historical match data, including team performance statistics, head to head records, home and away advantage, player availability, and recent form, is collected and pre processed to build a structured dataset suitable for supervised learning. Exploratory data analysis is carried out to identify the features that most strongly influence match results, followed by feature engineering steps such as encoding categorical variables, handling missing values, and normalizing numerical attributes. The pre processed dataset is then split into training and testing subsets, and both Logistic Regression and Random Forest models are trained to classify outcomes as win, loss, or draw.

Logistic Regression is chosen for its interpretability and ability to model linear relationships between features and outcomes, while Random Forest is selected for its capability to capture complex, non linear interactions and reduce overfitting through ensemble learning. The performance of both models is evaluated and compared using metrics such as accuracy, precision, recall, F1 score, and confusion matrices, allowing a clear assessment of their relative strengths and weaknesses. Experimental results indicate that the Random Forest model generally achieves higher predictive accuracy compared to Logistic Regression, owing to its ability to model complex feature interactions, although Logistic Regression offers valuable interpretability for understanding the contribution of individual factors to match outcomes.

The project also discusses the limitations of the current approach, including data availability and the inherent unpredictability of sporting events, and suggests directions for future work such as incorporating real time data, advanced ensemble techniques, and deep learning models. Overall, this study demonstrates the practical value of machine learning techniques in sports analytics and provides a foundation for building more accurate and reliable match outcome prediction systems for the sports and entertainment domain.

2. INTRODUCTION:

Sports events attract massive global attention, and the ability to accurately predict match outcomes has significant value for fans, analysts, media organizations, and the sports betting industry. Traditional prediction methods relied heavily on expert opinion and simple statistical averages, which often fail to capture the complex, dynamic factors that influence a game's result. With the growing availability of detailed match and player data, machine learning offers a data driven alternative that can identify hidden patterns and relationships beyond human intuition.

This project focuses on applying Logistic Regression and Random Forest algorithms to predict match outcomes based on historical performance data. Logistic Regression provides a simple, interpretable baseline, while Random Forest, an ensemble of decision trees, is capable of modeling more intricate, non linear relationships in the data. By comparing these two approaches, the project aims to determine which technique offers better predictive performance for this domain, while also highlighting the trade off between model interpretability and accuracy. The insights gained from this study can support better decision making in team strategy, fan engagement, and sports media analytics.

3. SOFTWARE REQUIREMENTS:

The development and evaluation of the match outcome prediction system rely on a Python based machine learning ecosystem. The table below summarizes the software components used across data preprocessing, model building, evaluation, and reporting stages of the project.

Component	Specification / Requirement
Operating System	Windows 10 / 11, Linux (Ubuntu 20.04+), or macOS
Programming Language	Python 3.9 or above
IDE / Notebook Environment	Jupyter Notebook, Google Colab, VS Code, or PyCharm
Core Libraries	NumPy, Pandas (data handling and preprocessing)
Machine Learning Libraries	Scikit-learn (Logistic Regression, Random Forest, model evaluation)
Visualization Libraries	Matplotlib, Seaborn (exploratory data analysis and result plots)
Data Source / Storage	CSV / Excel datasets, or SQL database for historical match data
Version Control	Git and GitHub for source code management
Documentation Tools	Microsoft Word / LaTeX for report preparation
Web Framework (optional)	Flask / Streamlit for deploying the prediction model as a demo application

4.HARDWARE REQUIREMENTS:

Since Logistic Regression and Random Forest are computationally lightweight compared to deep learning models, the project does not require specialized high performance hardware. A standard personal computer or laptop meeting the following specifications is sufficient for data preprocessing, model training, and evaluation.

Component	Specification / Requirement
Processor	Intel Core i5 (8th Gen) / AMD Ryzen 5 or higher
RAM	8 GB minimum (16 GB recommended for faster model training)
Storage	256 GB SSD or higher, with at least 5 GB free space for datasets and libraries
Graphics	Integrated graphics sufficient (dedicated GPU not mandatory for these algorithms)
Internet Connectivity	Required for downloading datasets, libraries, and using cloud notebooks
Display	Standard monitor with minimum 1366 x 768 resolution
Input Devices	Keyboard and mouse

